Feb 23, 2026

Predicting `beam_x`

Simulated Data

- Momentum (p): $p = p_{\text{nom}} X, \quad X \sim \mathcal{U}(0.8, 1.2)$ MeV
- In-Plane Angle (ip): $\text{ip} \sim \mathcal{U}(-2.0°, 2.0°)$
- Out-of-Plane Angle (oop): $\text{oop} \sim \mathcal{U}(-5.5°, 5.5°)$
- Beam hit position on the target:

$$
\begin{aligned}
&R = 20 \text{ mm}, \\
&U \sim \mathcal{U}(0,1), \quad \phi \sim \mathcal{U}(0, 2\pi), \\
&r = R\sqrt{U}, \\
&x = r\cos\phi = R\sqrt{U}\,\cos\phi, \\
&y = r\sin\phi = R\sqrt{U}\,\sin\phi, \\
&z = 0
\end{aligned}
$$

Simulated Data

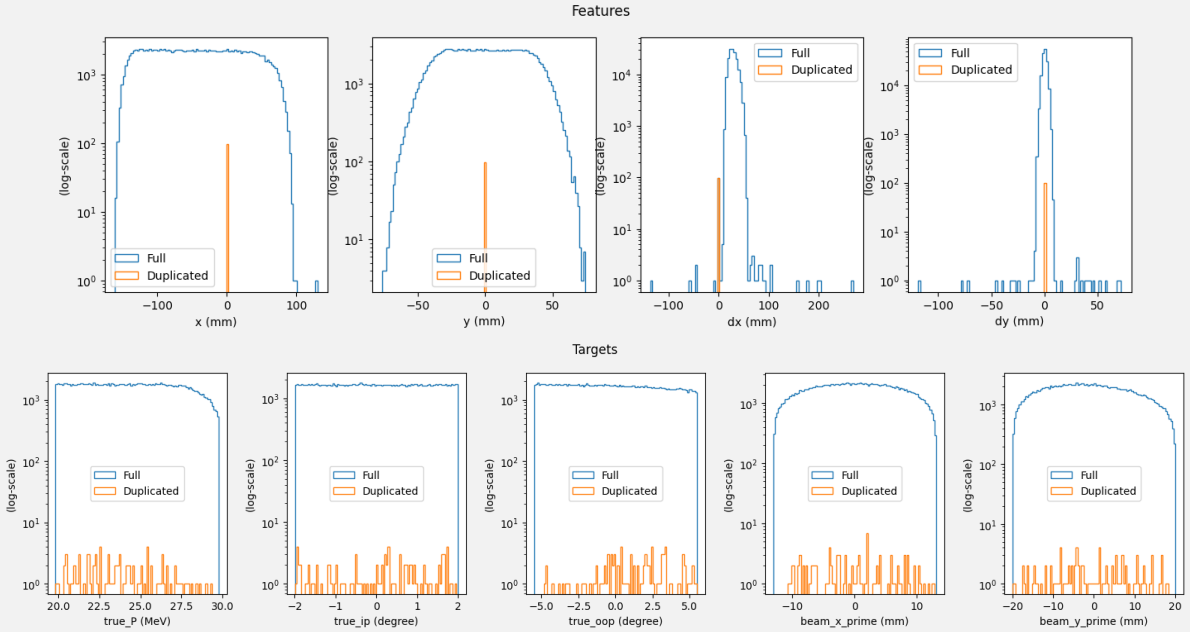
- Momentum (p): $p = p_{\text{nom}} X, \quad X \sim \mathcal{U}(0.8, 1.2)$ MeV
- In-Plane Angle (ip): $\text{ip} \sim \mathcal{U}(-2.0°, 2.0°)$
- Out-of-Plane Angle (oop): $\text{oop} \sim \mathcal{U}(-5.5°, 5.5°)$
- Beam hit position on the target:

$$
\begin{aligned}
R &= 20 \text{ mm}, \\
U &\sim \mathcal{U}(0,1), \quad \phi \sim \mathcal{U}(0, 2\pi), \\
r &= R\sqrt{U}, \\
x &= r\cos\phi = R\sqrt{U}\,\cos\phi, \\
y &= r\sin\phi = R\sqrt{U}\,\sin\phi, \\
z &= 0
\end{aligned}
$$

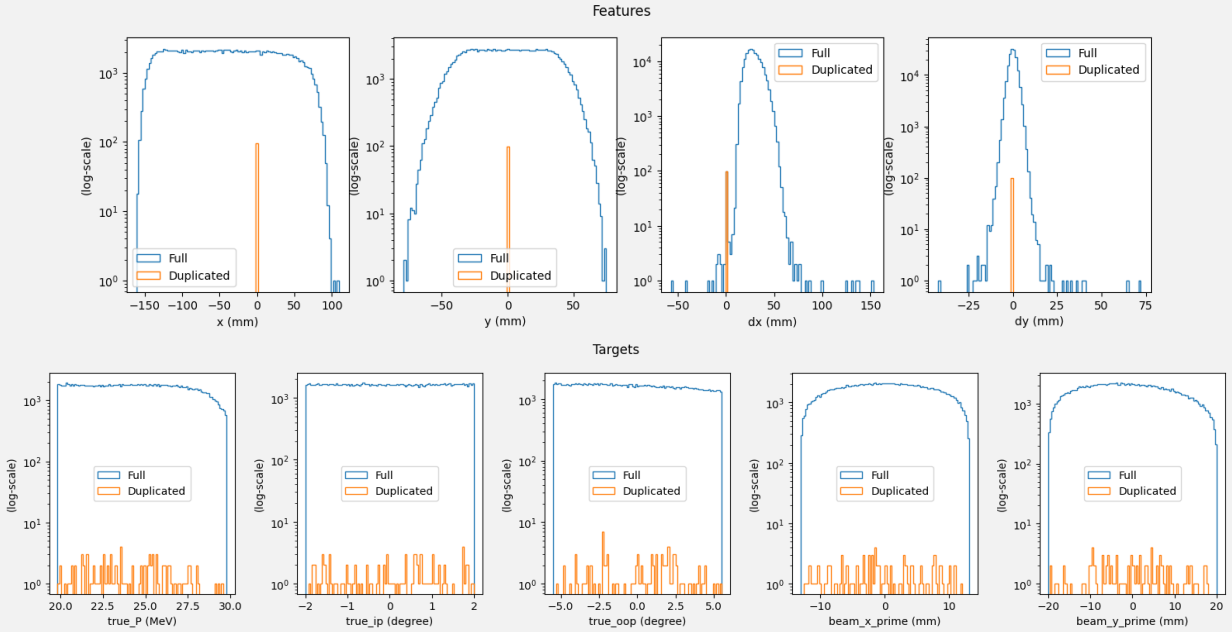
- To switch off multiple scattering, we run:

```
/process/inactivate msc
/process/inactivate eBrem
/process/inactivate CoulombScat
```

Distribution of Data: Features and Targets

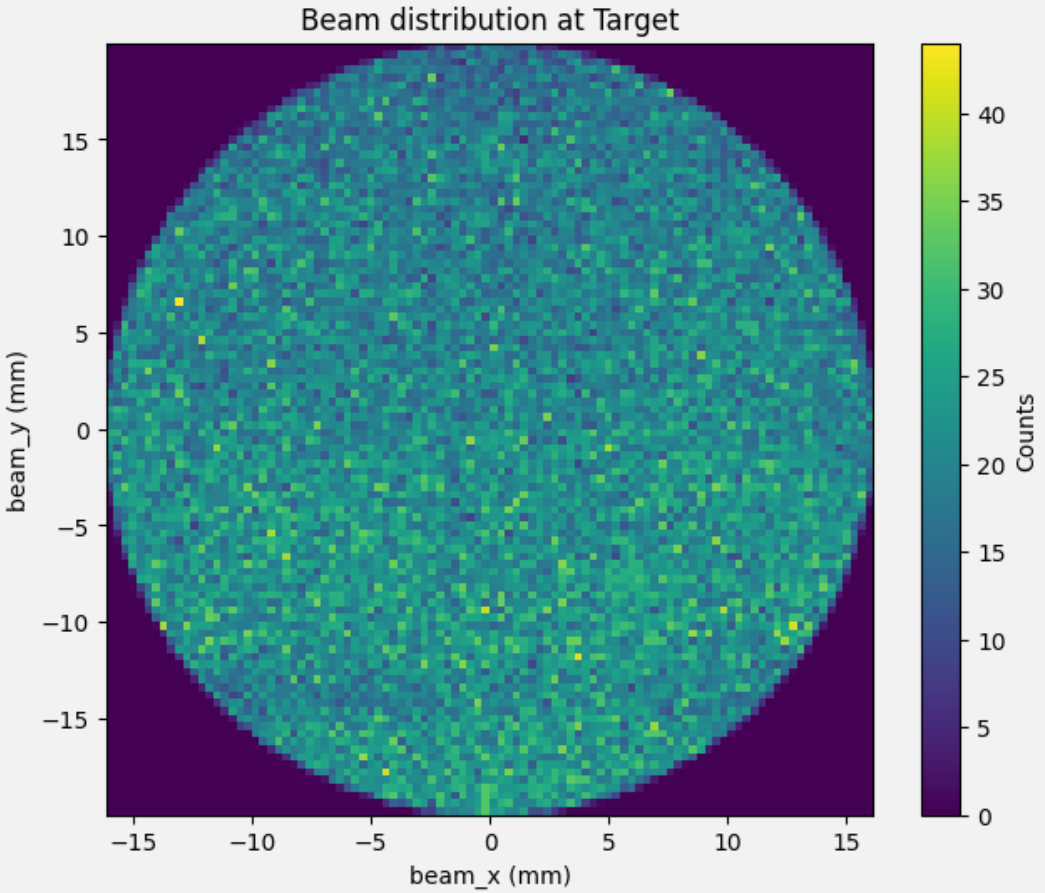


Multiple Scattering: OFF

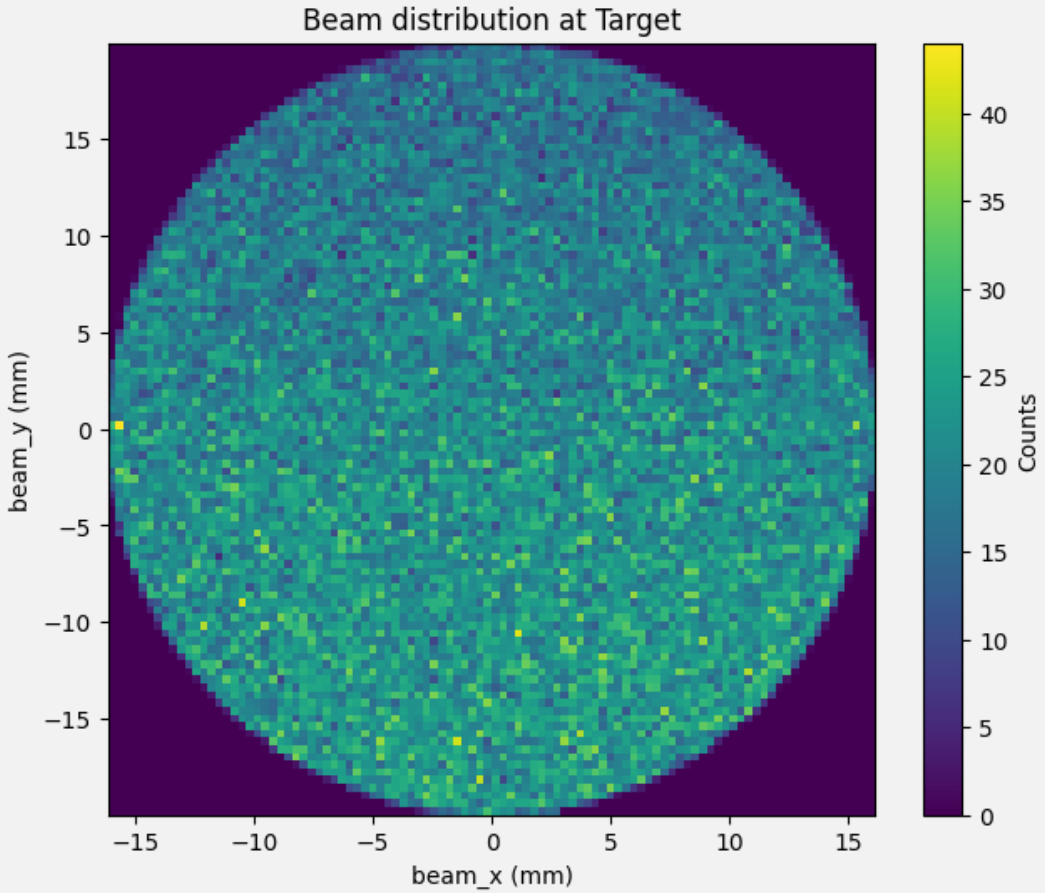


Multiple Scattering: ON

Multiple Scattering: OFF

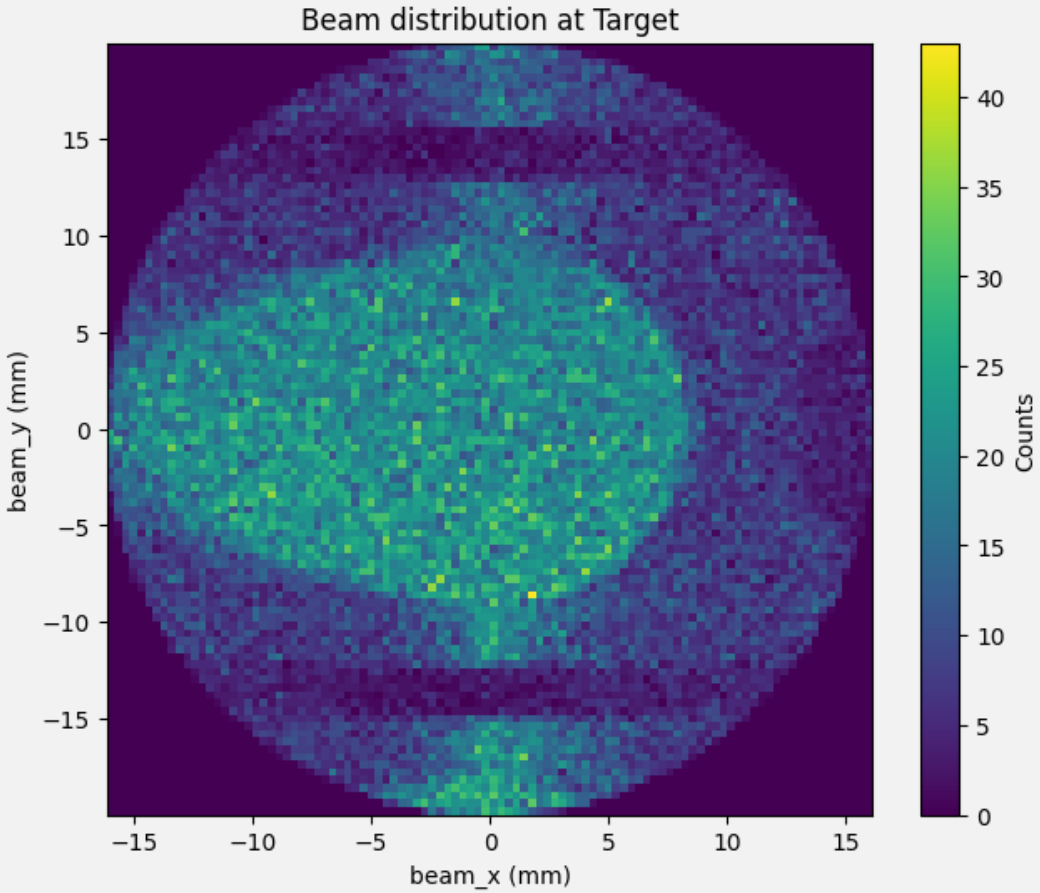


With Target Ladder

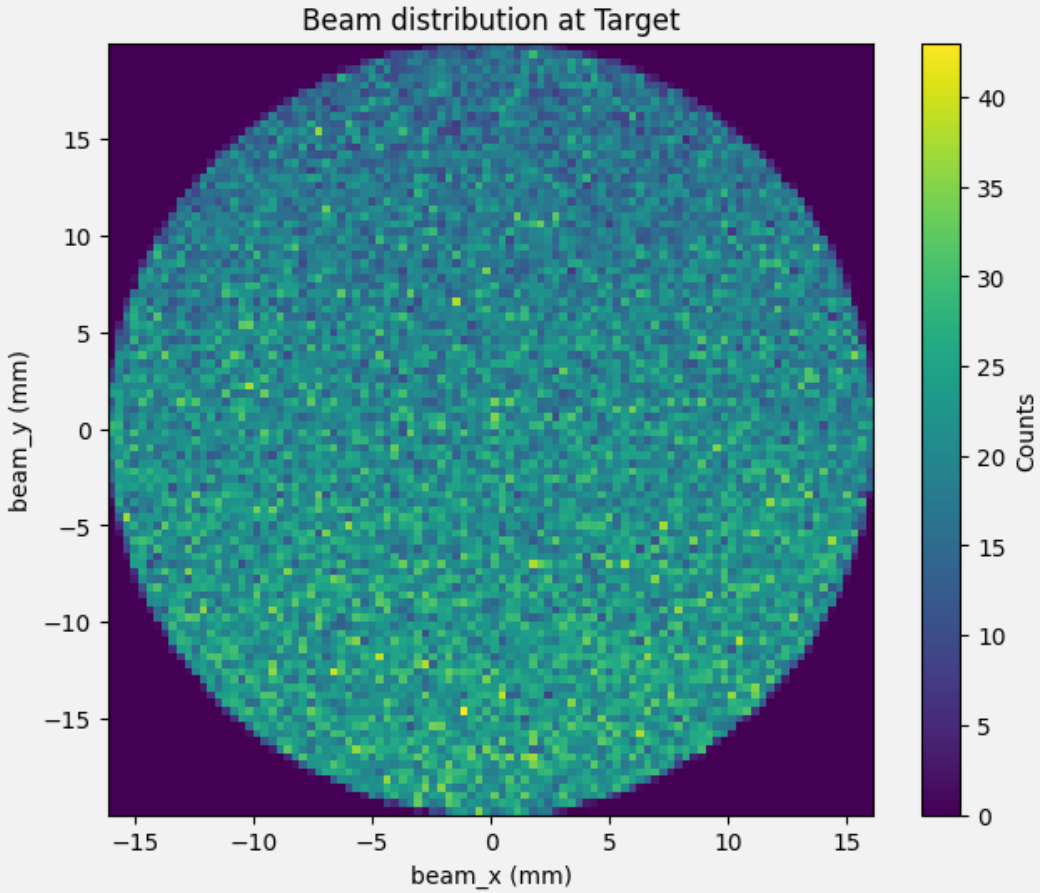


Without Target Ladder

Multiple Scattering: ON



With Target Ladder

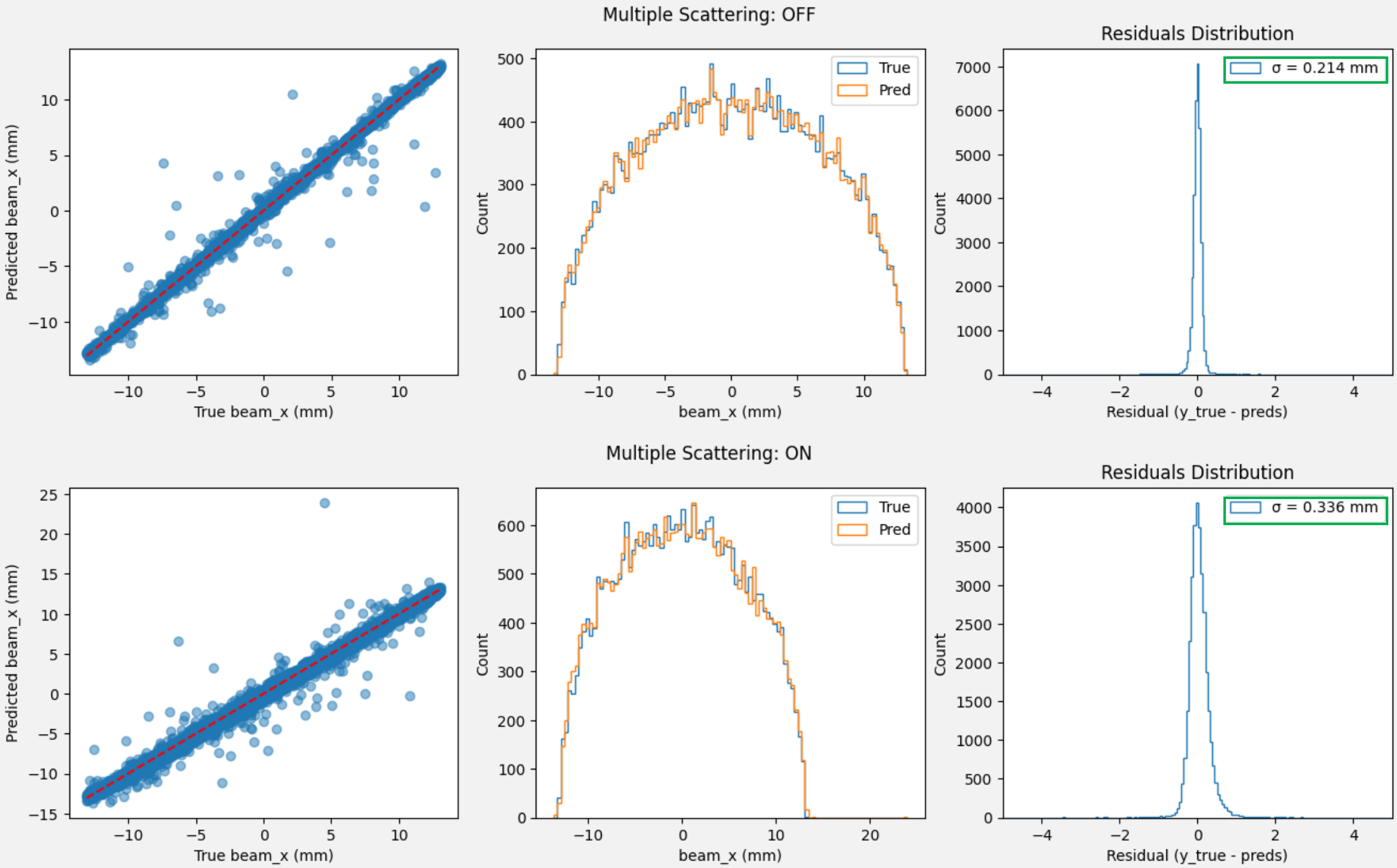


Without Target Ladder

Is `beam_x` recoverable?

- We build an *Oracle Model*, which uses the true values of target-related features that are not available at test time.
- These features include: momentum (p), in-plane angle (ip), out-of-plane angle (oop), path (path), and time (time).
- Therefore, the model learns the following mapping:

$$f : \underbrace{(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}, \mathrm{p}, \mathrm{ip}, \mathrm{oop}, \mathrm{path}, \mathrm{time})}_{\text{Input features}} \longrightarrow \underbrace{\mathrm{beam_x}}_{\text{Target}}$$


Multiple Scattering: OFF
Predicted beam_x (mm)
True beam_x (mm)
Count
beam_x (mm)
True
Pred
Residuals Distribution
σ = 0.214 mm
Residual (y_true - preds)
Multiple Scattering: ON
Predicted beam_x (mm)
True beam_x (mm)
Count
beam_x (mm)
True
Pred
Residuals Distribution
σ = 0.336 mm
Residual (y_true - preds)

Model 1: Multi-Task Neural Network with Target Weighting

Objective: Learn a multi-task regression mapping

$$f_\theta : \mathbb{R}^4 \to \mathbb{R}^6$$

$$\mathbf{u} = (x, y, dx, dy) \;\longmapsto\; f_\theta(\mathbf{u}) = (P,\ ip,\ oop,\ path,\ time,\ \mathbf{beam_x}')$$

with enhanced emphasis on accurate prediction of $\mathbf{beam_x}'$.

Architecture

- Input features (standardized): x, y, dx, dy
- Fully connected network:

$$4 \to 128 \to 64 \to 32 \to 6$$

- ReLU activations + Dropout (0.1)
- Adam optimizer + weight decay (10^{-5})

Model 1: Multi-Task Neural Network with Target Weighting

Objective: Learn a multi-task regression mapping

$$f_\theta : \mathbb{R}^4 \to \mathbb{R}^6$$

$$\mathbf{u} = (x, y, dx, dy) \longmapsto f_\theta(\mathbf{u}) = (P,\ ip,\ oop,\ path,\ time,\ \mathbf{beam_x}')$$

with enhanced emphasis on accurate prediction of $\mathbf{beam_x}'$.

Architecture

- Input features (standardized): x, y, dx, dy
- Fully connected network:

$$4 \to 128 \to 64 \to 32 \to 6$$

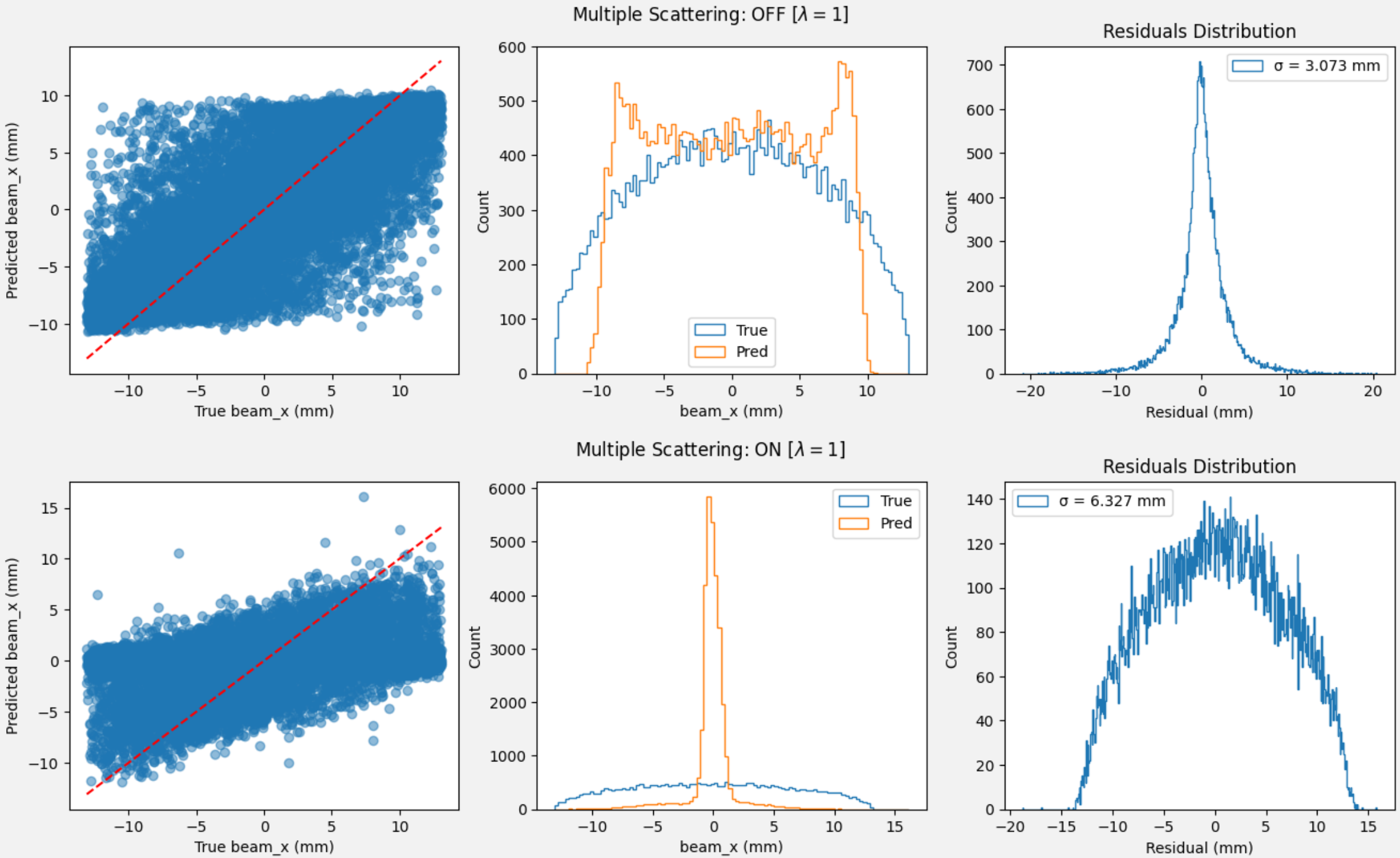
- ReLU activations + Dropout (0.1)
- Adam optimizer + weight decay (10^{-5})

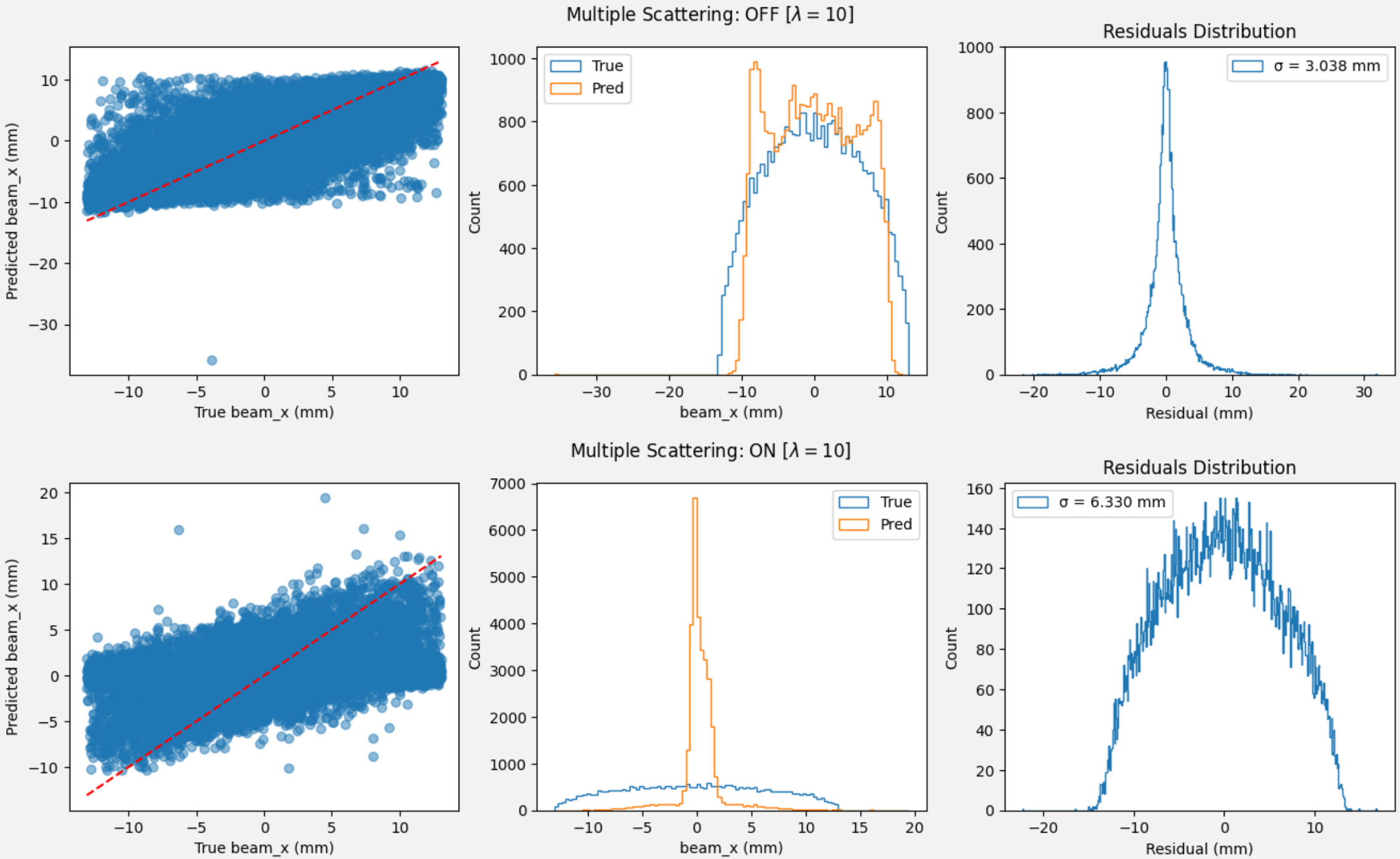
Weighted Multi-Task Loss

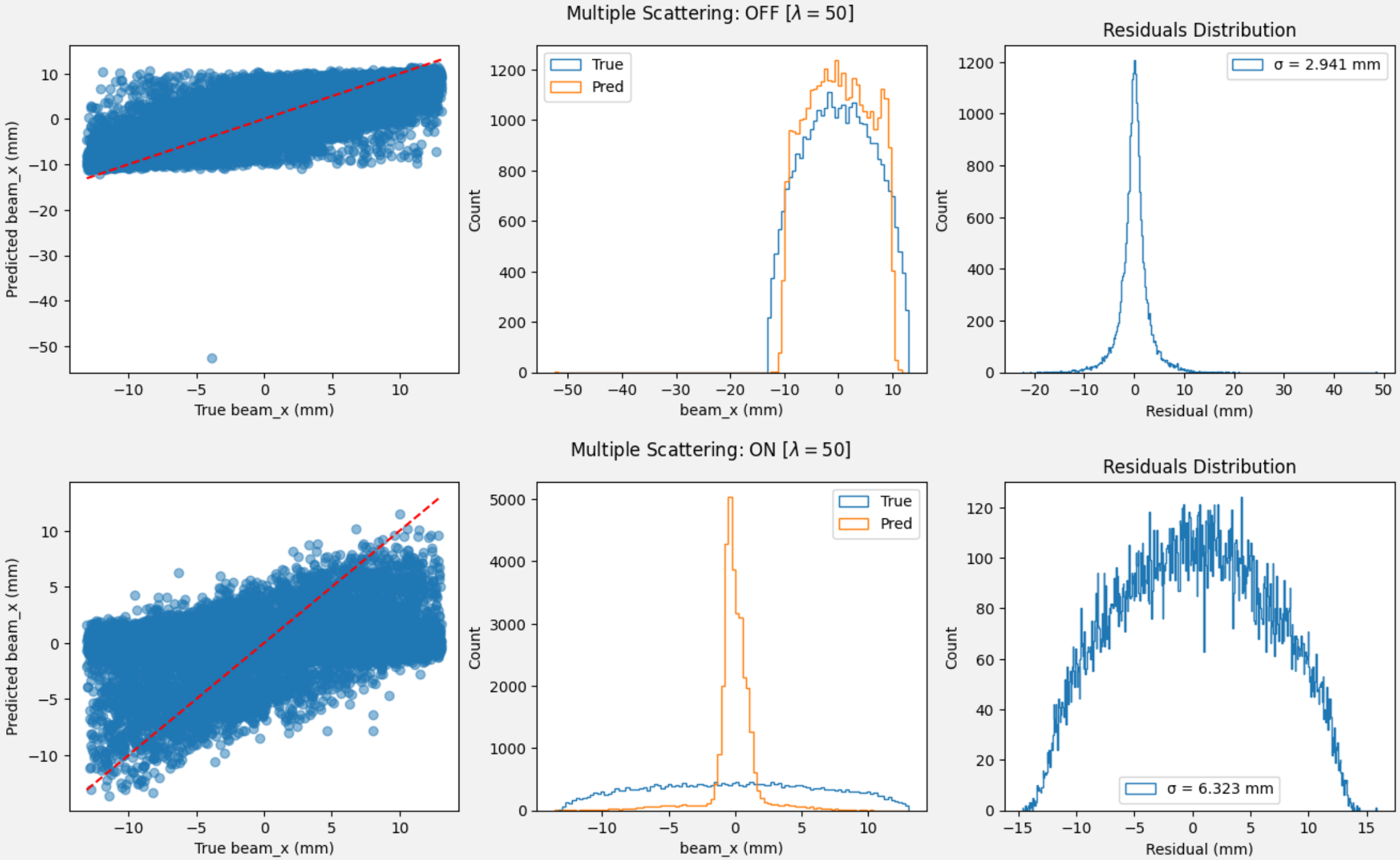
$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^{N} \left[\sum_{k=1}^{5} (\hat{y}_{ik} - y_{ik})^2 + \lambda \left(\hat{y}_{i,\mathbf{beam_x'}} - y_{i,\mathbf{beam_x'}} \right)^2 \right]$$

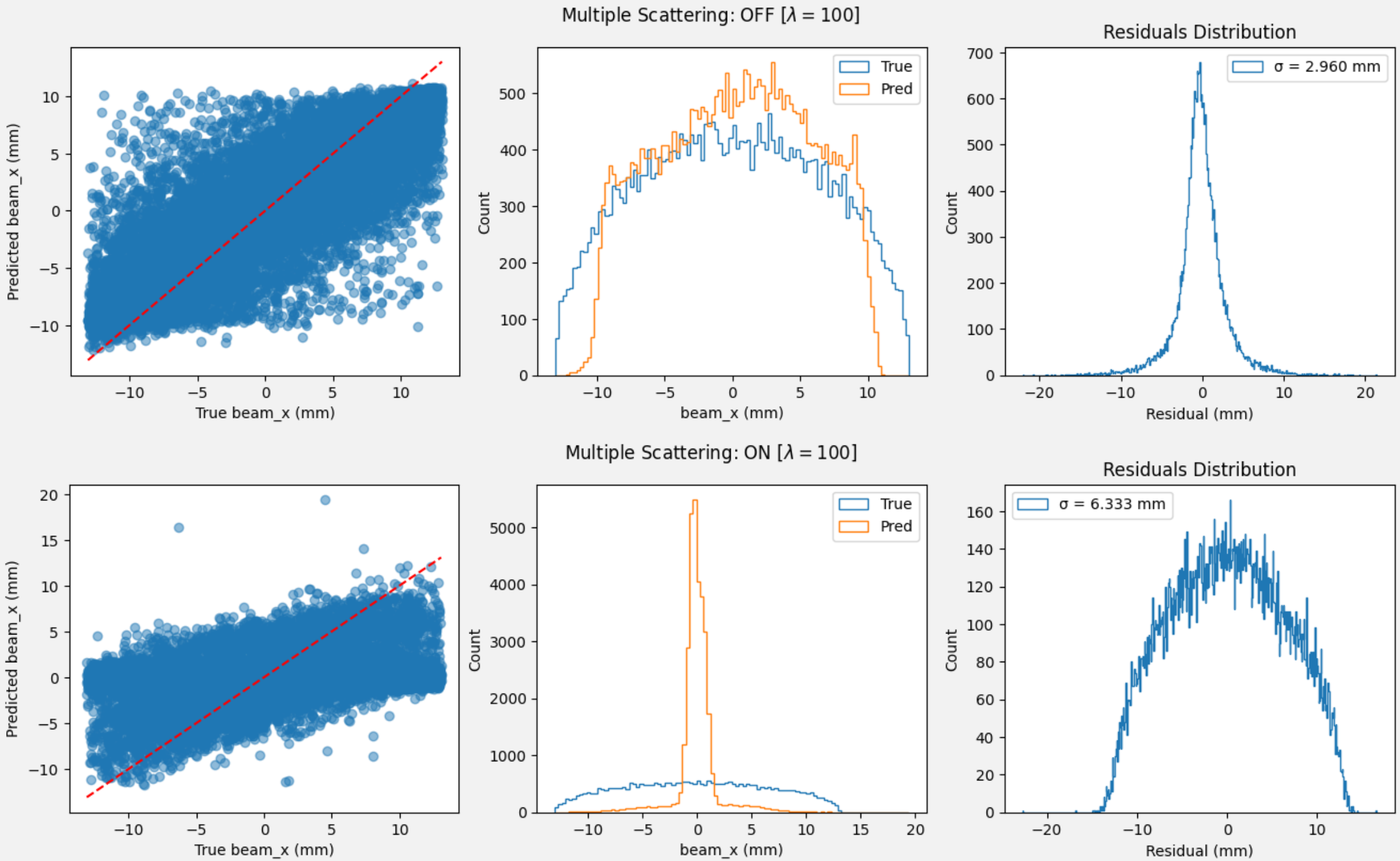
$$\boxed{\lambda = \text{weighing factor}}$$

Prioritizing resolution of $\mathbf{beam_x}'$ while retaining shared feature learning.


Multiple Scattering: OFF [λ = 1]
Predicted beam_x (mm)
True beam_x (mm)
Count
beam_x (mm)
True
Pred
Residuals Distribution
σ = 3.073 mm
Residual (mm)
Multiple Scattering: ON [λ = 1]
Predicted beam_x (mm)
True beam_x (mm)
Count
beam_x (mm)
True
Pred
Residuals Distribution
σ = 6.327 mm
Residual (mm)


Multiple Scattering: OFF [λ = 10]
Predicted beam_x (mm)
True beam_x (mm)
True
Pred
Count
beam_x (mm)
Residuals Distribution
σ = 3.038 mm
Count
Residual (mm)
Multiple Scattering: ON [λ = 10]
Predicted beam_x (mm)
True beam_x (mm)
True
Pred
Count
beam_x (mm)
Residuals Distribution
σ = 6.330 mm
Count
Residual (mm)


Multiple Scattering: OFF [λ = 50]
Predicted beam_x (mm)
True beam_x (mm)
True
Pred
Count
beam_x (mm)
Residuals Distribution
σ = 2.941 mm
Count
Residual (mm)
Multiple Scattering: ON [λ = 50]
Predicted beam_x (mm)
True beam_x (mm)
True
Pred
Count
beam_x (mm)
Residuals Distribution
σ = 6.323 mm
Count
Residual (mm)


Multiple Scattering: OFF [λ = 100]
Predicted beam_x (mm)
True beam_x (mm)
True
Pred
Count
beam_x (mm)
Residuals Distribution
σ = 2.960 mm
Count
Residual (mm)
Multiple Scattering: ON [λ = 100]
Predicted beam_x (mm)
True beam_x (mm)
True
Pred
Count
beam_x (mm)
Residuals Distribution
σ = 6.333 mm
Count
Residual (mm)

Model 2: Two Step Prediction

Step 1. Randomly partition the dataset into two equally sized, disjoint subsets.
Let $\mathcal{I}_1, \mathcal{I}_2 \subset \{1, \ldots, 2n\}$ be disjoint index sets such that

$$\mathcal{I}_1 \cup \mathcal{I}_2 = \{1, \ldots, 2n\}, \qquad |\mathcal{I}_1| = |\mathcal{I}_2| = n.$$

Then define

$$\mathcal{D}_1 = \{(\mathbf{x}_i, y_i)\}_{i \in \mathcal{I}_1}, \tag{1}$$
$$\mathcal{D}_2 = \{(\mathbf{x}_i, y_i)\}_{i \in \mathcal{I}_2}. \tag{2}$$

Model 2: Two Step Prediction

Step 1. Randomly partition the dataset into two equally sized, disjoint subsets. Let $\mathcal{I}_1, \mathcal{I}_2 \subset \{1, \ldots, 2n\}$ be disjoint index sets such that

$$\mathcal{I}_1 \cup \mathcal{I}_2 = \{1, \ldots, 2n\}, \qquad |\mathcal{I}_1| = |\mathcal{I}_2| = n.$$

Then define

$$\mathcal{D}_1 = \{(\mathbf{x}_i, y_i)\}_{i \in \mathcal{I}_1}, \tag{1}$$
$$\mathcal{D}_2 = \{(\mathbf{x}_i, y_i)\}_{i \in \mathcal{I}_2}. \tag{2}$$

Step 2. Train Models 1–5 on $\mathcal{D}_1$ to learn the mappings

$$\begin{aligned}
f_1 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{p}, \\
f_2 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{ip}, \\
f_3 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{oop}, \\
f_4 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{time}, \\
f_5 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{path}.
\end{aligned}$$

Model 2: Two Step Prediction

Step 1. Randomly partition the dataset into two equally sized, disjoint subsets. Let $\mathcal{I}_1, \mathcal{I}_2 \subset \{1, \ldots, 2n\}$ be disjoint index sets such that

$$\mathcal{I}_1 \cup \mathcal{I}_2 = \{1, \ldots, 2n\}, \qquad |\mathcal{I}_1| = |\mathcal{I}_2| = n.$$

Then define

$$\mathcal{D}_1 = \{(\mathbf{x}_i, y_i)\}_{i \in \mathcal{I}_1}, \tag{1}$$
$$\mathcal{D}_2 = \{(\mathbf{x}_i, y_i)\}_{i \in \mathcal{I}_2}. \tag{2}$$

Step 2. Train Models 1–5 on $\mathcal{D}_1$ to learn the mappings

$$\begin{aligned}
f_1 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{p}, \\
f_2 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{ip}, \\
f_3 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{oop}, \\
f_4 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{time}, \\
f_5 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{path}.
\end{aligned}$$

Step 3. Use Models 1–5 to generate predictions on $\mathcal{D}_2$:

$$\begin{aligned}
\widehat{\mathrm{p}} &= f_1(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}), \\
\widehat{\mathrm{ip}} &= f_2(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}), \\
\widehat{\mathrm{oop}} &= f_3(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}), \\
\widehat{\mathrm{time}} &= f_4(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}), \\
\widehat{\mathrm{path}} &= f_5(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy})
\end{aligned}$$

for all samples in $\mathcal{D}_2$.

Model 2: Two Step Prediction

Step 1. Randomly partition the dataset into two equally sized, disjoint subsets. Let $\mathcal{I}_1, \mathcal{I}_2 \subset \{1, \ldots, 2n\}$ be disjoint index sets such that

$$\mathcal{I}_1 \cup \mathcal{I}_2 = \{1, \ldots, 2n\}, \qquad |\mathcal{I}_1| = |\mathcal{I}_2| = n.$$

Then define

$$\mathcal{D}_1 = \{(\mathbf{x}_i, y_i)\}_{i \in \mathcal{I}_1}, \tag{1}$$
$$\mathcal{D}_2 = \{(\mathbf{x}_i, y_i)\}_{i \in \mathcal{I}_2}. \tag{2}$$

Step 2. Train Models 1–5 on $\mathcal{D}_1$ to learn the mappings

$$\begin{aligned}
f_1 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{p}, \\
f_2 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{ip}, \\
f_3 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{oop}, \\
f_4 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{time}, \\
f_5 &: (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}) \longmapsto \mathrm{path}.
\end{aligned}$$

Step 3. Use Models 1–5 to generate predictions on $\mathcal{D}_2$:

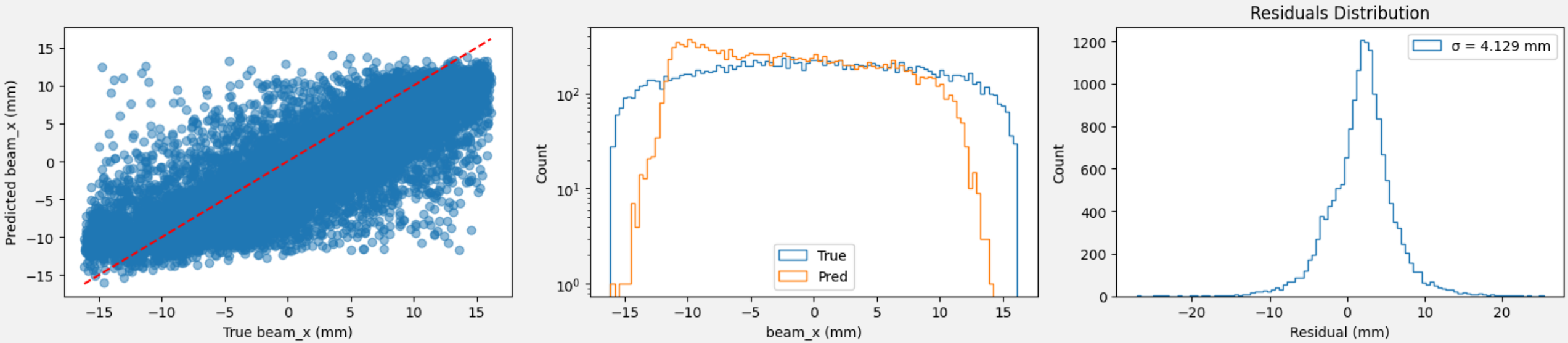
$$\begin{aligned}
\widehat{\mathrm{p}} &= f_1(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}), \\
\widehat{\mathrm{ip}} &= f_2(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}), \\
\widehat{\mathrm{oop}} &= f_3(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}), \\
\widehat{\mathrm{time}} &= f_4(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}), \\
\widehat{\mathrm{path}} &= f_5(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy})
\end{aligned}$$

for all samples in $\mathcal{D}_2$.

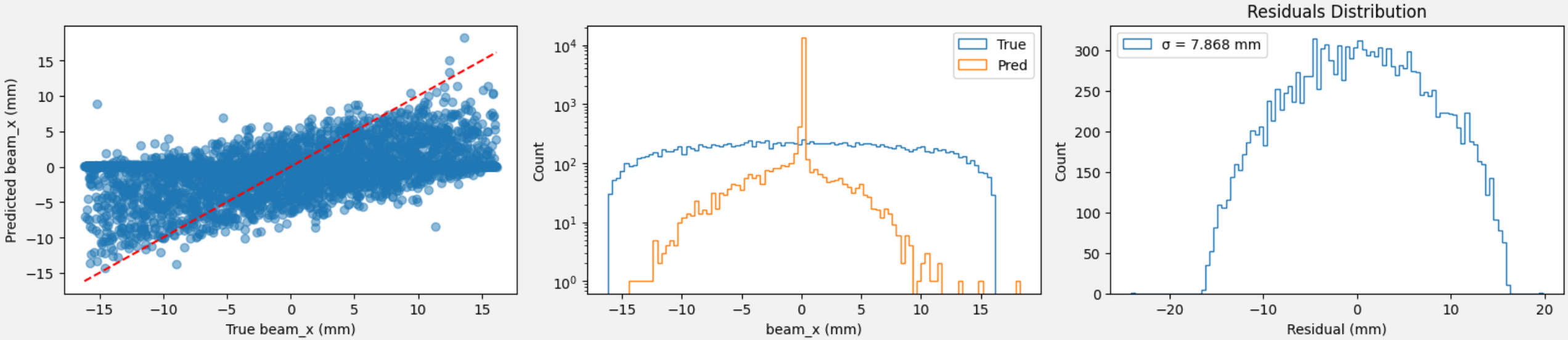
Step 4. Train Model 2 on $\mathcal{D}_2$ to learn the mapping

$$g : (\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}, \widehat{\mathrm{p}}, \widehat{\mathrm{ip}}, \widehat{\mathrm{oop}}, \widehat{\mathrm{time}}, \widehat{\mathrm{path}}) \longmapsto \mathrm{beam_x}.$$

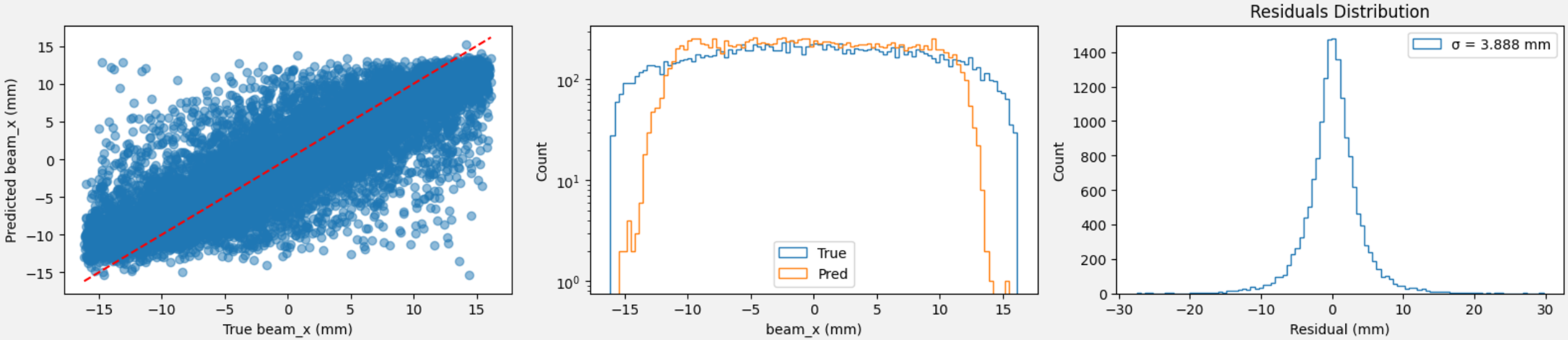
Multiple Scattering: OFF [Features: ['x', 'y', 'dx', 'dy', 'p_pred', 'ip_pred', 'oop_pred', 'path_pred', 'time_pred']]



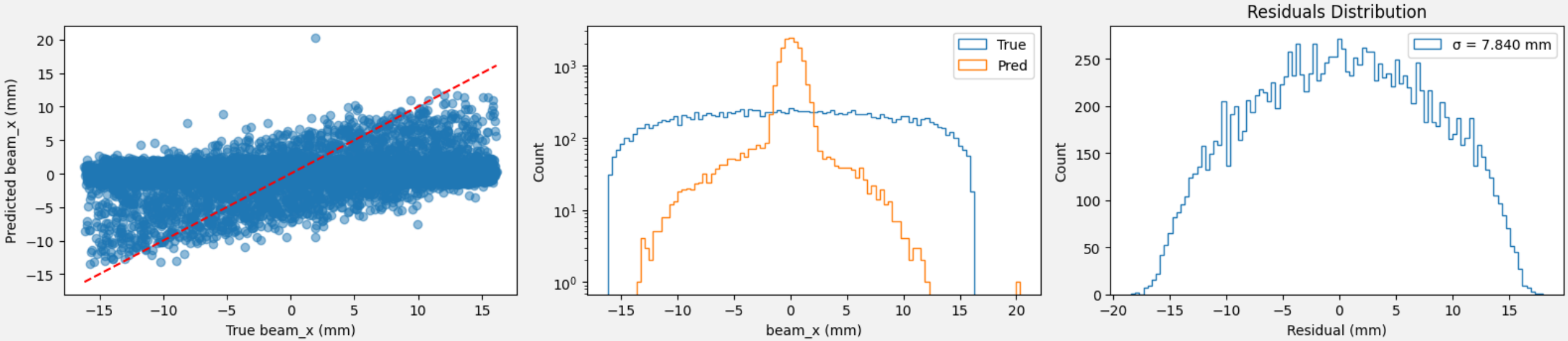
Multiple Scattering: ON [Features: ['x', 'y', 'dx', 'dy', 'p_pred', 'ip_pred', 'oop_pred', 'path_pred', 'time_pred']]



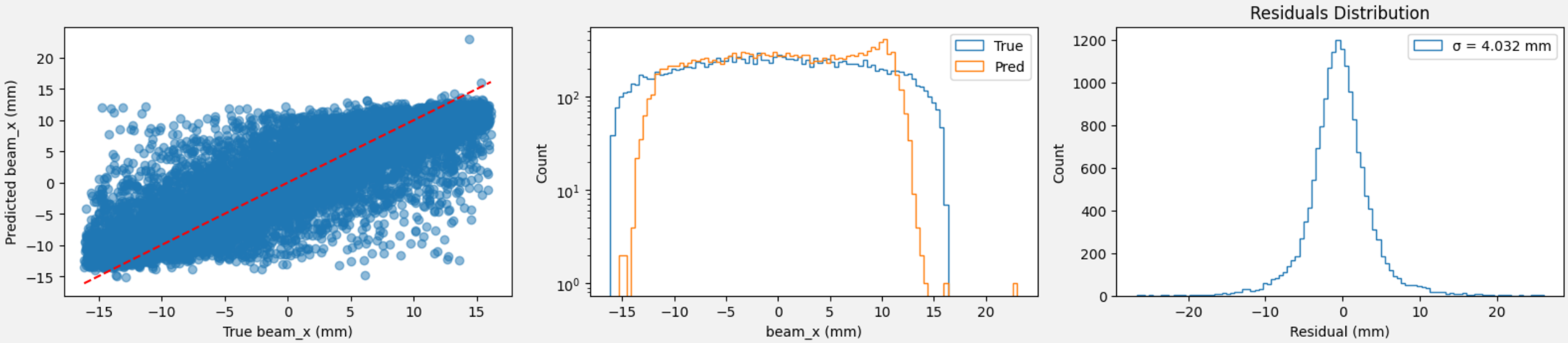
Multiple Scattering: OFF [Features: ['x', 'y', 'dx', 'dy', 'p_pred', 'ip_pred', 'oop_pred', 'time_pred']]



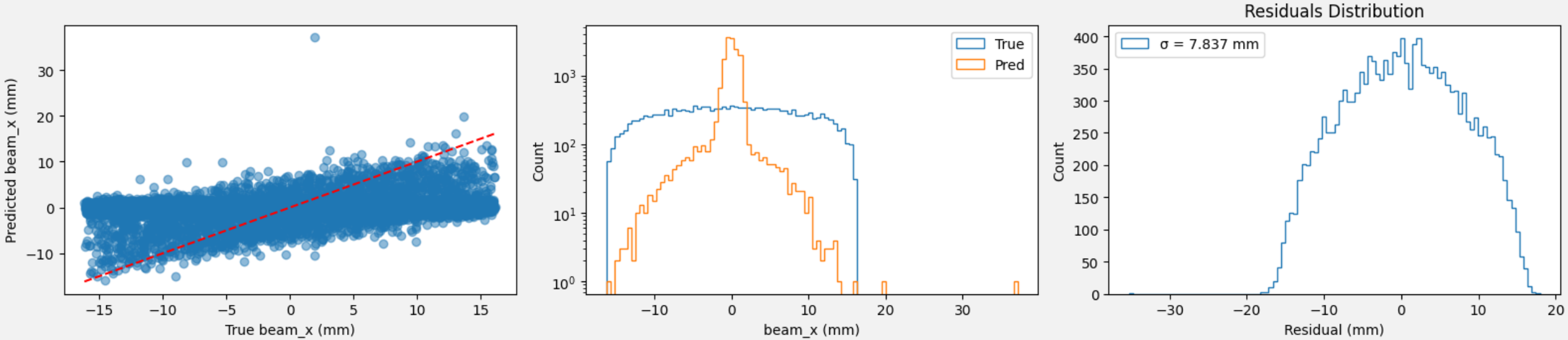
Multiple Scattering: ON [Features: ['x', 'y', 'dx', 'dy', 'p_pred', 'ip_pred', 'oop_pred', 'time_pred']]



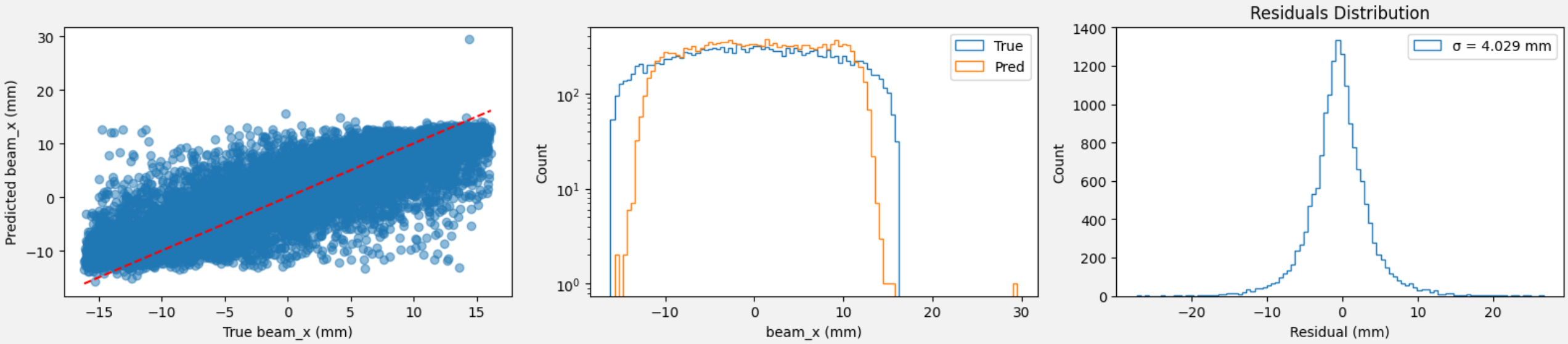
Multiple Scattering: OFF [Features: ['x', 'y', 'dx', 'dy', 'p_pred', 'ip_pred']]



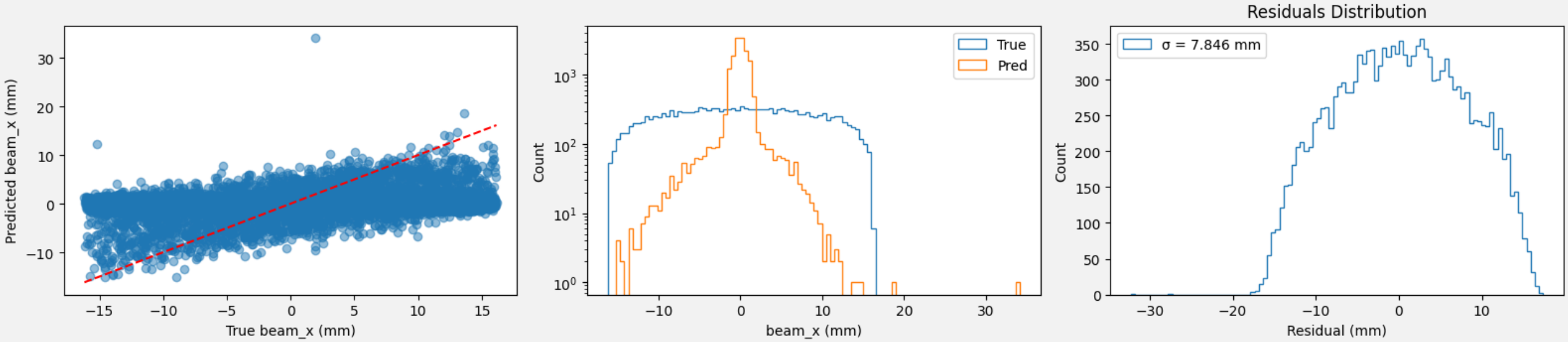
Multiple Scattering: ON [Features: ['x', 'y', 'dx', 'dy', 'p_pred', 'ip_pred']]



Multiple Scattering: OFF [Features: ['x', 'y', 'dx', 'dy', 'ip_pred']]



Multiple Scattering: ON [Features: ['x', 'y', 'dx', 'dy', 'ip_pred']]



Model 3: Multitask Gaussian

What Does the Network Learn?

Step 1 — Shared Representation
From the inputs

$$(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}),$$

the network learns a nonlinear latent representation

$$h = \phi(x) \in \mathbb{R}^{32}.$$

Model 3: Multitask Gaussian

What Does the Network Learn?

Step 1 — Shared Representation
From the inputs

$$(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}),$$

the network learns a nonlinear latent representation

$$h = \phi(x) \in \mathbb{R}^{32}.$$

Step 2 — Deterministic Predictions
From h, the model predicts

$$(\mathrm{p},\ \mathrm{ip},\ \mathrm{oop},\ \mathrm{path},\ \mathrm{time})$$

using standard regression.

Model 3: Multitask Gaussian

What Does the Network Learn?

Step 1 — Shared Representation
From the inputs

$$(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}),$$

the network learns a nonlinear latent representation

$$h = \phi(x) \in \mathbb{R}^{32}.$$

Step 2 — Deterministic Predictions
From h, the model predicts

$$(\mathrm{p},\ \mathrm{ip},\ \mathrm{oop},\ \mathrm{path},\ \mathrm{time})$$

using standard regression.

Step 3 — Probabilistic Prediction
For beam_x, the model predicts

$$\mu(x), \quad \sigma(x),$$

assuming

$$\mathrm{beam_x} \sim \mathcal{N}(\mu(x), \sigma^2(x)).$$

Model 3: Multitask Gaussian

What Does the Network Learn?

Step 1 — Shared Representation
From the inputs

$$(\mathrm{x}, \mathrm{y}, \mathrm{dx}, \mathrm{dy}),$$

the network learns a nonlinear latent representation

$$h = \phi(x) \in \mathbb{R}^{32}.$$

Step 2 — Deterministic Predictions
From h, the model predicts

$$(\mathrm{p},\ \mathrm{ip},\ \mathrm{oop},\ \mathrm{path},\ \mathrm{time})$$

using standard regression.

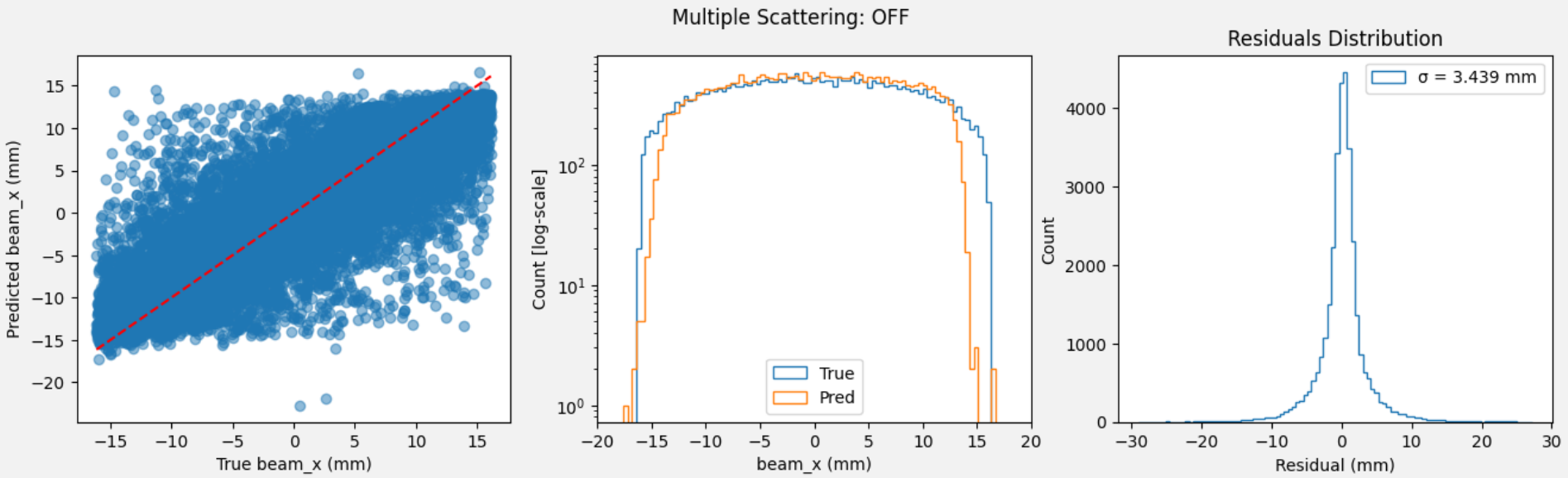
Step 3 — Probabilistic Prediction
For beam_x, the model predicts

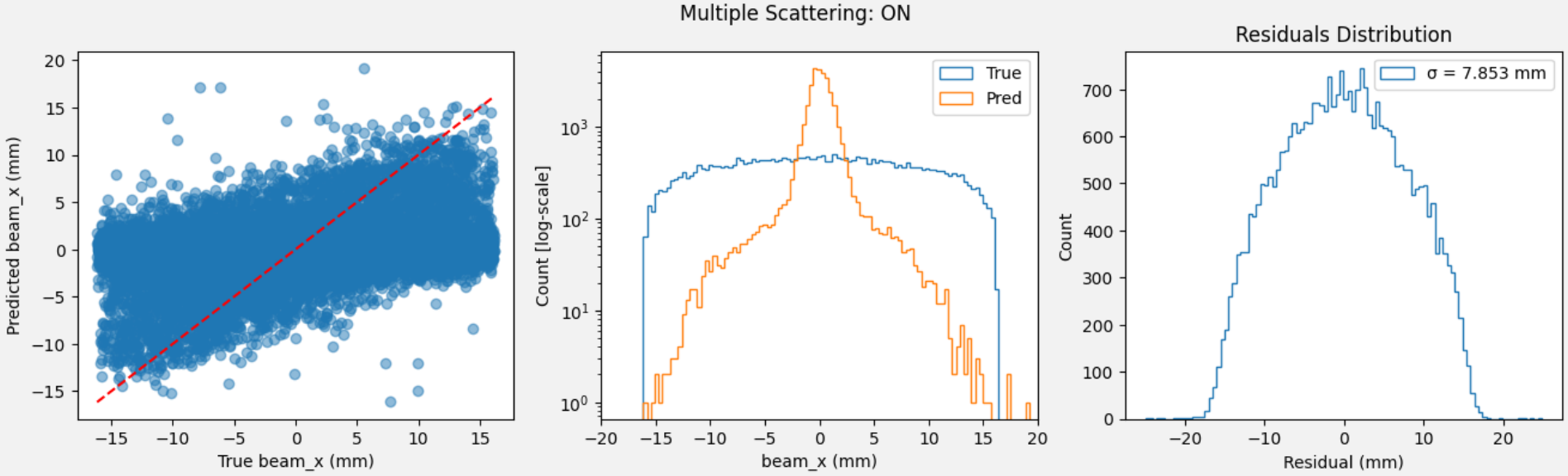
$$\mu(x), \quad \sigma(x),$$

assuming

$$\mathrm{beam_x} \sim \mathcal{N}(\mu(x), \sigma^2(x)).$$

Summary:
The network learns a shared nonlinear representation of the inputs, from which it predicts five deterministic quantities and one Gaussian-distributed quantity with learned uncertainty.


Multiple Scattering: OFF
Predicted beam_x (mm)
True beam_x (mm)
Count [log-scale]
beam_x (mm)
True
Pred
Residuals Distribution
σ = 3.439 mm
Count
Residual (mm)


Multiple Scattering: ON
Predicted beam_x (mm)
True beam_x (mm)
Count [log-scale]
beam_x (mm)
True
Pred
Residuals Distribution
σ = 7.853 mm
Count
Residual (mm)